



Andrés Gerardo García Marcano

Mechanical Engineering Student
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LinkedIn Profile

PROFILE

Mechanical engineering student with experience in modeling, analysis of mechanical and structural components, as well as drafting and design process development. Recognized for strong skills in planning, teamwork, professional ethics, precision, and responsibility. Proficient in FEA simulation tools, CAD modeling, and the application of geometric dimensioning and tolerancing (GD&T). Interested in mechanical design and simulation, focusing on the development of efficient and optimized systems for the aerospace, transportation, and infrastructure sectors.

EDUCATION

Degree	University	CGPA	Year	Location
Bachelor's Degree in Mechanical Engineering	EAFIT University	4.11/5.00	July 2021 - Present	Colombia

EXPERIENCE

- Proworks Colombia S.A.S.** Jul. 2023 - Present
The company specializes in lifeline systems. Medellín, Colombia
 - CAD modeling, static analysis, and technical documentation for lifeline systems
 - 4 lifeline system design and analysis projects | 300+ meters installed
 - 2 structural projects (metal & wood)
 - High-precision component drafting

ACADEMIC PROJECTS

- Machine Design 2** Jan. 2025 - Present
Design and construction of an electric vehicle inspired by 1960s Formula 1 cars.
- Machine Design 1** Jan. 2024 - May. 2024
Development of a CNC router with three degrees of freedom and linear actuation.
- Finite Element Method** Jul. 2023 - Nov. 2023
Verification and validation of a finite element model of a 2-ton hydraulic crane.
- Mechanisms** Jan. 2023 - Jun. 2023
Design and construction of a human-powered vehicle applying concepts learned during the course.

SKILLS

- Softwares:** SolidWorks, Inventor
- Programming:** Matlab, Python, Arduino, Latex
- Office software:** Microsoft 365
- Soft Skills:** Time management, Interdisciplinary collaboration, Problem-solving, Adaptability, Continuous learning

EXTRACURRICULAR ACTIVITIES

- Kratos Program** Jan. 2022 - Nov. 2024
EAFIT University Medellín, Colombia
 - Design Process Leader for an aluminum-built Human-Powered Vehicles (HPV) | 2024
 - Chassis Design & Manufacturing Leader for a Electric-Powered Vehicles (EPV) | 2023
 - Design Team Member for Developing a HPV conceptual design | 2022
- Undergraduate Teacher Assistant | Functional Analysis of Machines** Jan. 2025 - Present
EAFIT University Medellín, Colombia
 - Creation of guideline documentation for design processes with dimensional, kinematic, static, and dynamic analysis of mechanisms.
- Undergraduate Teacher Assistant | Technical Drawing** Jan. 2022 - May. 2024
EAFIT University Medellín, Colombia
 - In-class student support for doubt resolution and weekly workshop facilitation.

– Assignment review and feedback to enhance learning outcomes.

• **Student Representative for the Mechanical Engineering Undergraduate Program**

Jul. 2023 - Jul. 2024

EAFIT University

Medellín, Colombia

- Liaison between the program committee and students to support the undergraduate program and institution.
- Organized a SolidWorks competition with over 40 participants.

• **Turbomachinery Research Group**

Jan. 2023 - May. 2023

EAFIT University

Medellín, Colombia

- Study of an asynchronous induction machine operating as a motor and as a generator

CERTIFICATIONS

- **SolidWorks**: CSWA Mechanical Design | Mar. 2023 | [C-V4LEPGDZX7](#)

ACHIEVEMENTS & AWARDS

- **Inventiva Award for Best Teamwork**, Student Organization (OE)

Jun. 2024

- **Engineer of the Year**, Kratos Program

Nov. 2023

- **Pilot of the Year**, Kratos Program

Nov. 2024

MEMBERSHIP IN PROFESSIONAL ORGANIZATIONS

- **ASME**, American Society of Mechanical Engineers

2024

LANGUAGES

- **English B1**, Tracktest (CEFR) - 2023

- **Spanish**, Native